DAS 2026 · KEYNOTE

Conceptual Loops for Enterprise Document AI

Reduce uncertainty. Don’t waste experience.

Dr. Jordy Van Landeghem · Probably Approximately Human

whoami

- Research intern ORACLE (*Seq2Seq for Dialogue Modeling*) and NUANCE (*LM Algorithms*)
- Founding AI Research Engineer contract.fit [2017-2024]
- PhD project: *Intelligent Automation for AI-driven Document Understanding* KU LEUVEN
- Model Dev - GenAI - Agents Technical Lead INSTABASE [2024-…]
- Founded my own company Probably Approximately Human [2026]

📍 Belgium 🇧🇪

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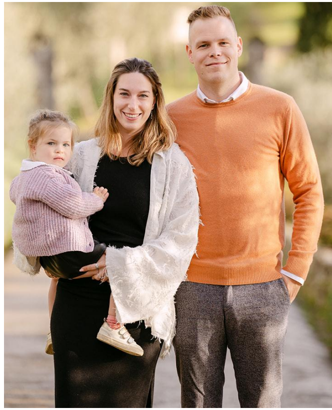
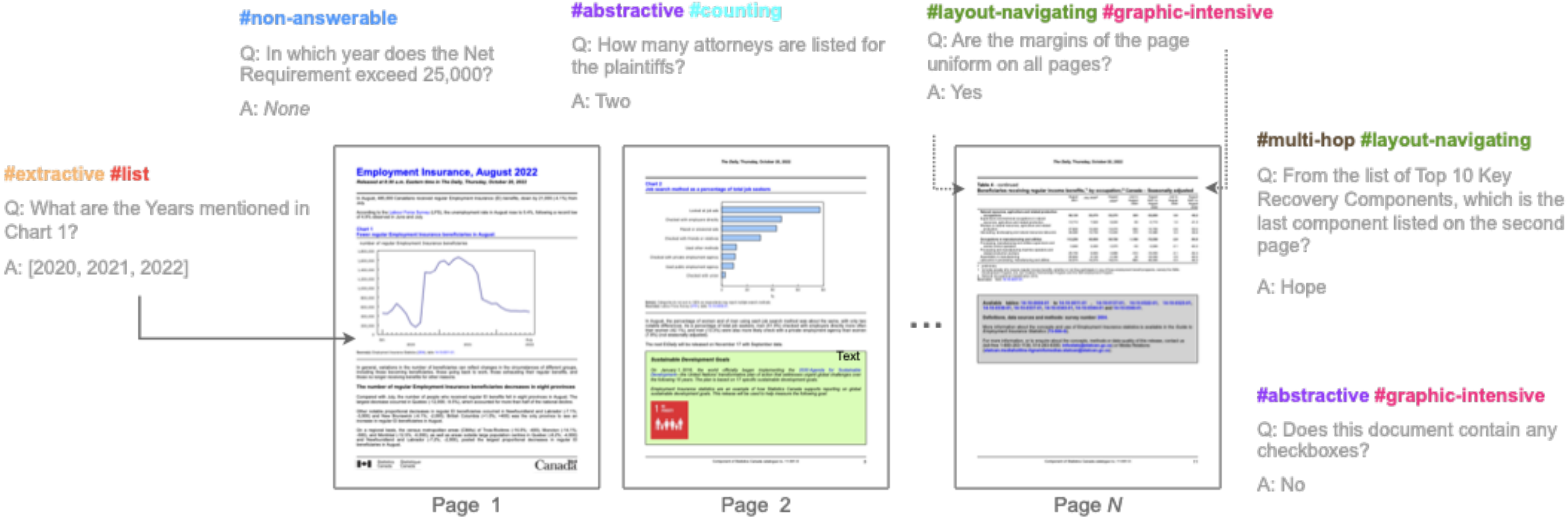
🏝️ 🤿 🗺️
👾 🍴



You might (hopefully) know me from …

*Document UnderstanDing of Everything (**DUDE**)* 😎



Jordy Van Landeghem, Rubén Tito, Łukasz Borchmann, Michał Pietruszka, Paweł Józiak, Rafał Powalski, Dawid Jurkiewicz, Mickaël Coustaty, Bertrand Ackaert, Ernest Valveny, Matthew Blaschko, Sien Moens, Tomasz Stanisławek. **Document Understanding Dataset and Evaluation (DUDE)**, [ICCV 2023]/ICDAR 2023

Or more recently….

***MADQA** / Strategic Navigation or Stochastic Search? How Agents and Humans Reason Over Document Collections*

Borchmann, Ł., Van Landeghem, J., Turski, M., Padarha, S., Kearns, R.O., Mahdi, A., Rogge, N., Fourrier, C., Han, S., Yao, H. and Llabrés, A., 2026. **Strategic Navigation or Stochastic Search? How Agents and Humans Reason Over Document Collections**. [ICML 2026] (oral)

Agenda

01

Reality checks from industry

From benchmark performance to production automation

Why f(x) = y stops being enough, and why a loop is needed at all

02

The inner loop

1. Representation
2. Search
3. Evidence
4. Action

Where agents can help, one uncertainty question per stage

03

The outer loop

- Experience & Feedback
- Memory (remember, promote, compile)
- Don't waste experience
- Open Research frontier

Section 01

Reality checks

Traditional DocAI vs. Modern Requirements

01

Problem statement

Two very different objectives

DOCUMENT UNDERSTANDING

$f(x) \rightarrow y$

Goal	benchmark performance
Unit	model / prediction
Accuracy	primary metric
Failure	wrong prediction
Logic	mostly learned
Learning	train → deploy

ENTERPRISE DOCUMENT AI

document → evidence → action → outcomes

Goal	automation + trust
Unit	end-to-end system
Accuracy	upper bound on automation
Failure	wrong + what happens next? (verify/trace/escalate)
Logic	prediction + rules + workflow
Learning	experience → feedback → improve

A benchmark asks, “Was the answer right?”
Production asks, “Can the process run reliably, cheaply, auditably, and at scale?”

What form does f(x) typically take?

VLMs

- extract via a single model call
- Closed API models & open-weight models

```python
from google.genai import types
with open(file_path, "rb") as f:
     data = f.read()
part = types.Part.from_bytes(data=data,
mime_type=mime_type)
```

f(x)_per_field.json

Schema Output (Trust Values)

```json
{
"value": "Extracted Data",
"confidence": 0.97,
"provenance": {
    "source_page": 1,
    "bounding_box": [120, 45, 200, 80]
  }
}
```

OBSERVED SILENT FAILURE

Is VLM(Document) → Structured Output sufficient?

Verbalized confidence induces silent truncation & non-determinism

GRITS

78.4 → 96.0

after removing the continuous confidence request

Activity

Date	Description	Credits	Debits	Balance
1/2/2018	Direct Deposit PAYPAL TRANSFER TRANSFER	9.16		$4217.16
2/2/2018	LYFT *RIDE SUN 11AM	35.99		$4253.15
2/2/2018	STARBUCKS CARD RELOAD 800-782-7282 WA	731.19		$4984.34
2/2/2018	TARGET T-3201		-6.51	$4990.85
3/2/2018	Check Card Purchase UMA`S ISABELA, PR		-6.05	$4996.9
4/2/2018	ATM Withdrawal		-2.56	$4999.46
6/2/2018	ADY*SPOTIFY USA	7.22		$5006.68
7/2/2018	LYFT *RIDE WED 7PM		-62.11	$5068.79
8/2/2018	CVS/PHARMACY #		-17.15	$5085.94
8/2/2018	CENTRO DE JUGUETES Y A AGUADILL, PR	69.93		$5155.87
9/2/2018	SPRINT *WIRELESS 800-639-6111 KS		-19.52	$5175.39
9/2/2018	DIRECT DEP		-26.21	$5201.6
10/2/2018	SQC*ANISE POUR		-221.16	$5422.76
10/2/2018	0282 PLANET FITNESS SAN FRANCISCO CA	6.99		$5429.75
10/2/2018	APL* ITUNES.COM/BILL 866-712-7753 CA		-2.06	$5431.81
11/2/2018	BART-CLIPPER M	21.99		$5453.8
11/2/2018	Service Fee		-10.4	$5464.2
11/2/2018	SALLY BEAUTY #1954 SAN FRANCISCO CA		-20.48	$5484.68
11/2/2018	MOBILE_CK_DEPOSIT		-7.01	$5491.69
12/2/2018	TARGET T-3201	4.31		$5496.0
13/2/2018	APL* ITUNES.COM/BILL 866-712-7753 CA		-32.62	$5528.62
14/2/2018	SQ *BLUE BOTTLE COFFEE		-17.26	$5545.88
15/2/2018	APL* ITUNES.COM/BILL 866-712-7753 CA		-77.15	$5623.03
15/2/2018	Service Fee	843.98		$6467.01
15/2/2018	LYFT *RIDE FRI 9PM		-6.52	$6473.53
15/2/2018	APL* ITUNES.COM/BILL 866-712-7753 CA		-65.21	$6538.74
16/2/2018	PLANET FIT - CLUB FEES - PLANET FIT		-11.27	$6550.01
16/2/2018	CHEVRON/CSI-091000 SAN FRANCISCO, CA	5.29		$6555.3
16/2/2018	SQC*DONALD CAR	3.32		$6558.62
16/2/2018	LYFT *RIDE SAT 4AM		-29.46	$6588.08
17/2/2018	ATM Withdrawal		-44.35	$6632.43
17/2/2018	MOBILE_CK_DEPOSIT		-12.62	$6645.05
17/2/2018	Check Card Purchase	0.1		$6645.15

Ally Bank Member FDIC STMTCMB100 05/201

COMBINED CUSTOMER STATEMENT

17/2/2018	SAFEWAY STORE 07 SAN FRANCISCO, CA	609.07		$7254.22
19/2/2018	FLIPPERS GOURMET BUR SAN FRANCISCO CA		-23.58	$7277.8
19/2/2018	SALLY BEAUTY #1954 SAN FRANCISCO CA		-28.3	$7306.1
20/2/2018	Excessive Transactions Fee	57.74		$7363.84
21/2/2018	SAFEWAY STORE SAN FRANCISCO, CA		-8.54	$7372.38
21/2/2018	SLING TV, LLC 877-811-4788 CO		-26.58	$7398.96
21/2/2018	LYFT *RIDE SAT 11PM		-8.9	$7407.86
21/2/2018	Check Card Purchase UMA`S ISABELA, PR		-36.76	$7444.62
21/2/2018	Check Card Purchase CLARO SAN JUAN, PR		-18.24	$7462.86
21/2/2018	SLING TV, LLC 877-811-4788 CO		-339.39	$7802.25
22/2/2018	Direct Deposit PAYPAL TRANSFER		-14.63	$7816.88

self-judging every value while transcribing a long table created an escape hatch.

New problems to solve, stochastic control over loops

Enterprise Reliability Requirements

- **Automation** (adaptivity)
- **Defensible failure handling** (confidence/validations)
- **Auditability** (provenance)
- **Long context handling** (structured transcription)
- **Determinism** (stable behavior)
- **Continual learning** (production flywheel)

The Paradigm Shift

Reliable DocAI is built around stochastic intelligence.

The future belongs to systems that spend intelligence where it counts, adapt strategy under uncertainty and explain why we should trust the result.

For Researchers

Measurable questions across the loops

This paradigm opens critical new paths to explore representation, retrieval, evidence, decisions, and learning throughout the execution cycle.

DESIGNING RELIABLE AGENT LOOPS

The agent loop, inside the feedback loop

Before full autonomy — it's a reliable, engineered system.

FEEDBACK LOOP — THE SYSTEM LEARNS

AGENT LOOP — THE WORK HAPPENS

REPRESENTATION
What is here?

SEARCH
Where is it?

ACTION
What should I do?

EVIDENCE
Can I trust it?

Experience → Feedback → Change the system

Remember
Promote
Compile

Section 02

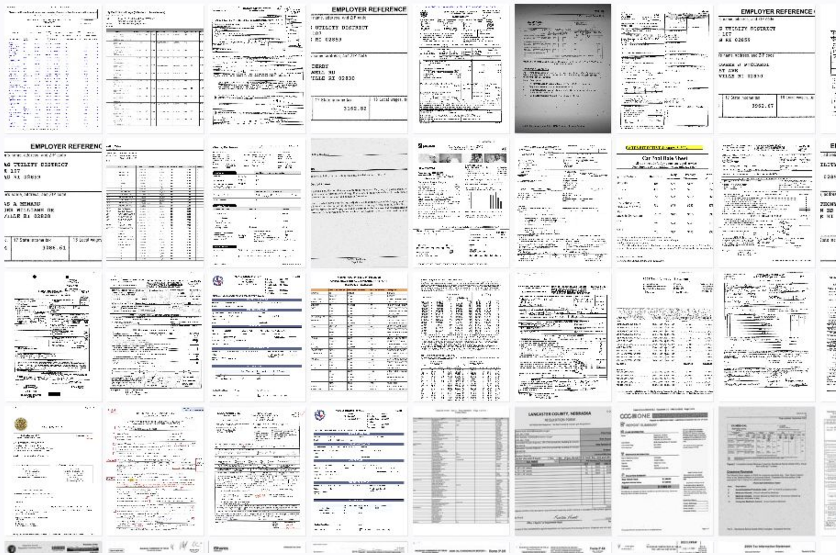
Inner Loop

REPRESENTATION

What is here?

What is actually in the document?

Can we represent the document faithfully?



ParseBench

<doclang/>

- text
- layout
- tables
- charts
- formatting
- reading order
- visual relations
- spatial semantics

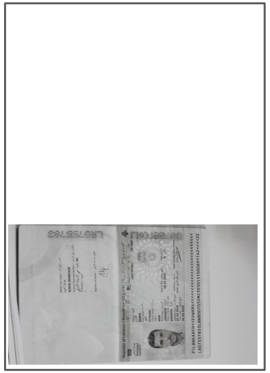
OBSERVATION

No method consistently dominates tables, charts, content faithfulness, formatting, and visual grounding.

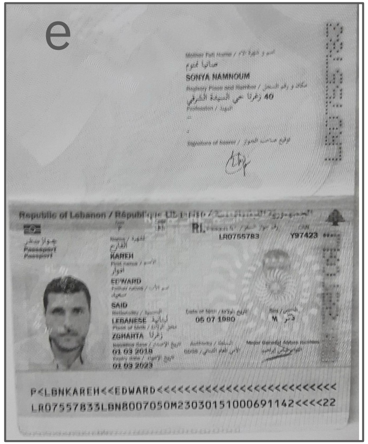
1 · REPRESENTATION - example

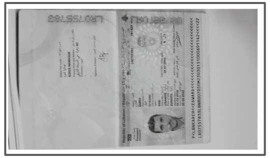
How would you read this complex document?

original



🔧upscale

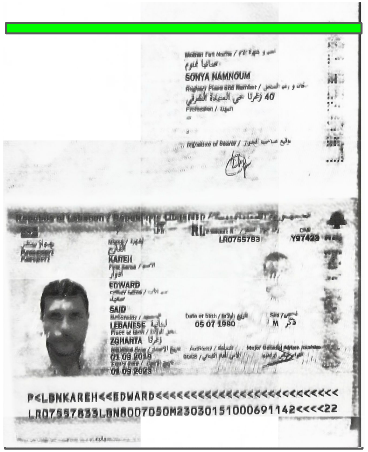




🔧crop

🔧🤖denoise

en, ar

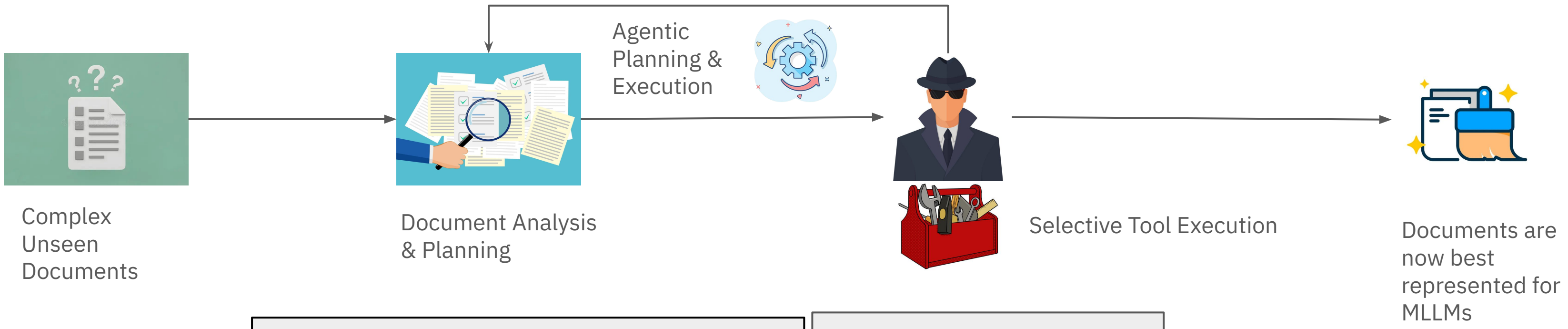


PaddleOCR

Google Cloud

Gemini Claude ChatGPT

So we built an Agent that *thinks with the documents*




```
Quality_assessment:
  - Overall_quality:  Medium
  - Text_readability: Fair
  - Image orientation:  Incorrect (rotated)
  - Color quality:  Washed out

Preprocessing recommendations:
  - Page Number 1:
    -  Apply minor  noise reduction to
       remove faint lines on the left
       margin (optional).
    -  Rotate the image 12° clockwise.
    -  Apply table extraction to capture
       transaction data.
    -  Decode postal barcode if its
       information is relevant.
```

Preprocessing Tools are applied on each page wherever required on **parallel.**

Not all pages or documents get processed, processing happens only where necessary

Tools at our disposal:
- Denoise Tool
- OCR Engines
- Table Recognition
- Deskew Tool
- Specialized VLMs
- …

Research question: Can agents optimize the accuracy / latency / cost frontier per document?

INSTABASE

Section 02

Inner Loop

SEARCH

Where is it?

Where is what I need?

WITHIN ONE DOCUMENT

100-page document

↓

find relevant regions

↓

crop

ACROSS A CORPUS

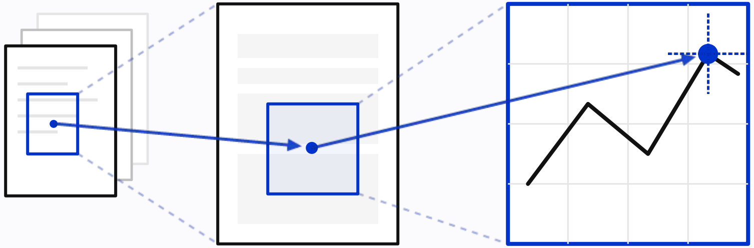
1M PDFs

↓

find relevant documents

↓

find relevant regions



01 / CORPUS
Documents

02 / REGION
Pages

03 / CROP
Target Evidence

> **Same fundamental operation. Different scale.**

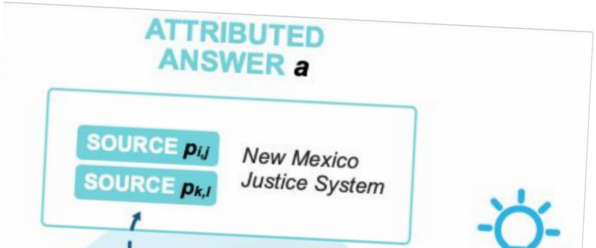
Strategic Navigation or Stochastic Search? How Agents and Humans Reason Over Document Collections

Łukasz Borchmann Jordy Van Landeghem Michał Turski Shreyansh Padarha
Ryan Othniel Kearns Adam Mahdi Niels Rogge Clémentine Fourrier
Siwei Han Huaxiu Yao Artemis Llabrés Yiming Xu Dimosthenis Karatzas
Hao Zhang Anupam Datta

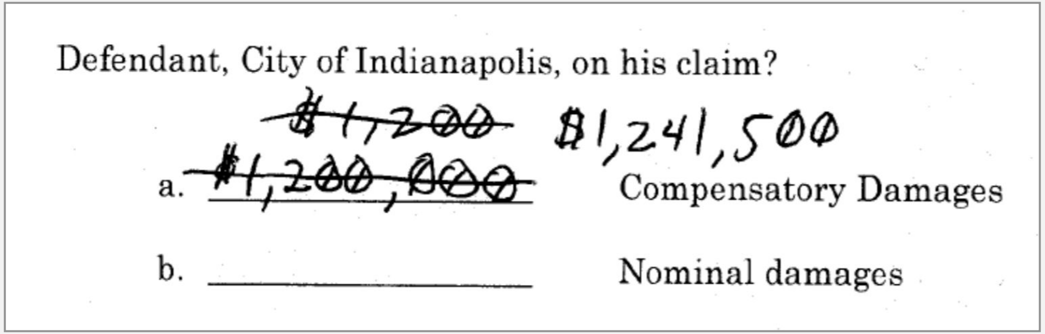
Snowflake/MADQA-Leaderboard </> Snowflake-Labs/MADQA OxRML/MADQA

Abstract

Multimodal agents offer a promising path to automating complex document-intensive workflows. Yet, a critical question remains: do these agents demonstrate genuine strategic reasoning, or merely stochastic trial-and-error search? To address this, we introduce MADQA, a benchmark of 2,250 human



MADQA Benchmark. Collection of Documents

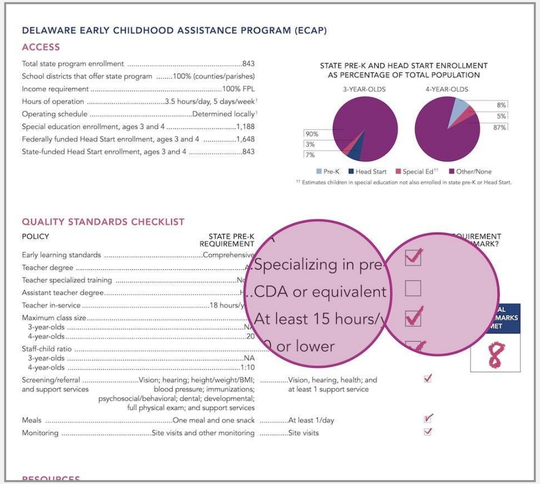


Defendant, City of Indianapolis, on his claim?

~~$1,200~~ $1,241,500

a. ~~$1,200,000~~ Compensatory Damages

b. ______________ Nominal damages



DELAWARE EARLY CHILDHOOD ASSISTANCE PROGRAM (ECAP)

ACCESS

Total state program enrollment 843
School districts that offer state program ....100% (counties/parishes)
Income requirement ....100% FPL
Hours of operation ....3.5 hours/day, 5 days/week[1]
Operating schedule ....Determined locally[1]
Special education enrollment, ages 3 and 4 1,188
Federally funded Head Start enrollment, ages 3 and 4 1,648
State-funded Head Start enrollment, ages 3 and 4 843

STATE PRE-K AND HEAD START ENROLLMENT AS PERCENTAGE OF TOTAL POPULATION

†† Estimates children in special education not also enrolled in state pre-K or Head Start.

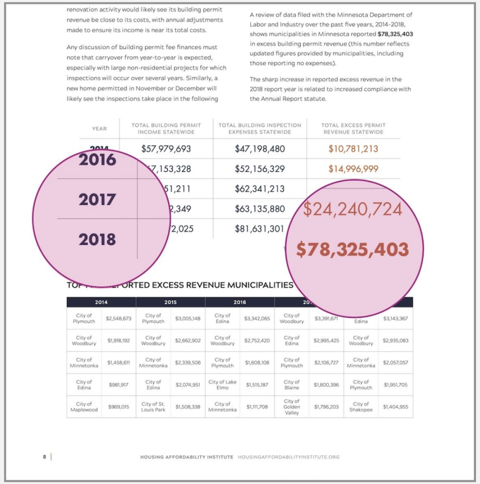
QUALITY STANDARDS CHECKLIST

POLICY	STATE PRE-K REQUIREMENT		REQUIREMENT MEETS BENCHMARK?
Early learning standards	Comprehensive		☑
Teacher degree		Specializing in pre	☐
Teacher specialized training	N	CDA or equivalent	☑
Assistant teacher degree	H	At least 15 hours/y	☑
Teacher in-service	18 hours/y	0 or lower	
Maximum class size			
3-year-olds	NA		
4-year-olds	20		
Staff-child ratio			
3-year-olds	NA		
4-year-olds	1:10		
Screening/referral and support services	Vision; hearing; height/weight/BMI; blood pressure; immunizations; psychosocial/behavioral; dental; developmental; full physical exam; and support services	Vision, hearing, health; and at least 1 support service	✓
Meals	One meal and one snack	At least 1/day	✓
Monitoring	Site visits and other monitoring	Site visits	✓

AL MARKS MET

8

RESOURCES



renovation activity would likely see its building permit revenue be close to its costs, with annual adjustments made to ensure its income is near its total costs.

Any discussion of building permit fee finances must note that carryover from year-to-year is expected, especially with large non-residential projects for which inspections will occur over several years. Similarly, a new home permitted in November or December will likely see the inspections take place in the following

A review of data filed with the Minnesota Department of Labor and Industry over the past five years, 2014-2018, shows municipalities in Minnesota reported **$78,325,403** in excess building permit revenue (this number reflects updated figures provided by municipalities, including those reporting no expenses).

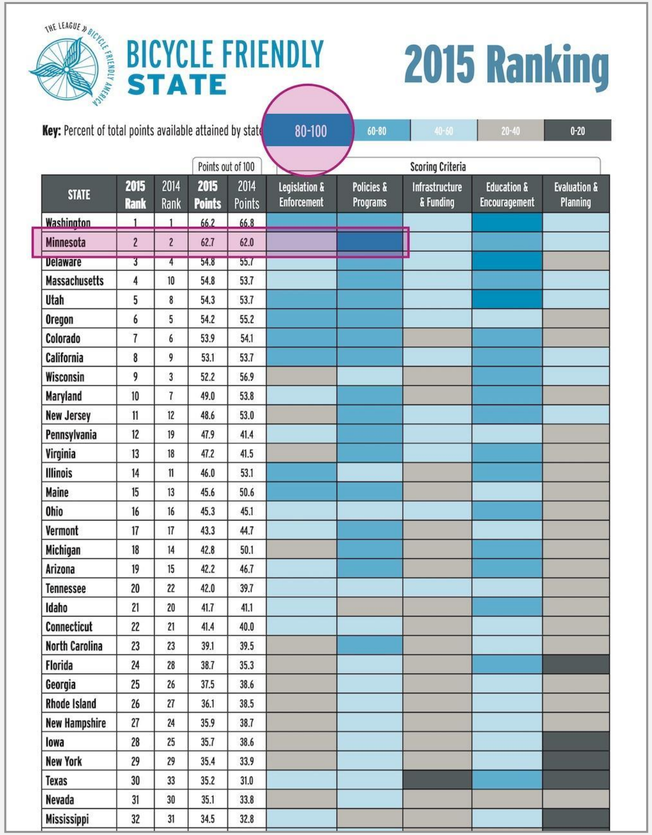
The sharp increase in reported excess revenue in the 2018 report year is related to increased compliance with the Annual Report statute.

YEAR	TOTAL BUILDING PERMIT INCOME STATEWIDE	TOTAL BUILDING INSPECTION EXPENSES STATEWIDE	TOTAL EXCESS PERMIT REVENUE STATEWIDE
2014	$57,979,693	$47,198,480	$10,781,213
2016	7,153,328	$52,156,329	$14,996,999
2017	51,211	$62,341,213	
2018	2,349	$63,135,880	$24,240,724
	2,025	$81,631,301	$78,325,403

TO ORTED EXCESS REVENUE MUNICIPALITIES

2014		2015		2016		20			
City of Plymouth	$2,548,673	City of Plymouth	$3,005,148	City of Edina	$3,342,065	City of Woodbury	$3,391,671	Edina	$3,143,367
City of Woodbury	$1,918,192	City of Woodbury	$2,662,902	City of Woodbury	$2,752,420	City of Edina	$2,995,425	City of Woodbury	$2,935,083
City of Minnetonka	$1,458,611	City of Minnetonka	$2,339,506	City of Plymouth	$1,608,108	City of Plymouth	$2,106,727	City of Minnetonka	$2,057,057
City of Edina	$981,917	City of Edina	$2,074,951	City of Lake Elmo	$1,515,187	City of Blaine	$1,800,396	City of Plymouth	$1,951,705
City of Maplewood	$969,015	City of St. Louis Park	$1,508,338	City of Minnetonka	$1,111,708	City of Golden Valley	$1,796,203	City of Shakopee	$1,404,955

8 | HOUSING AFFORDABILITY INSTITUTE HOUSINGAFFORDABILITYINSTITUTE.ORG



THE LEAGUE OF AMERICAN BICYCLISTS BICYCLE FRIENDLY AMERICA

BICYCLE FRIENDLY STATE

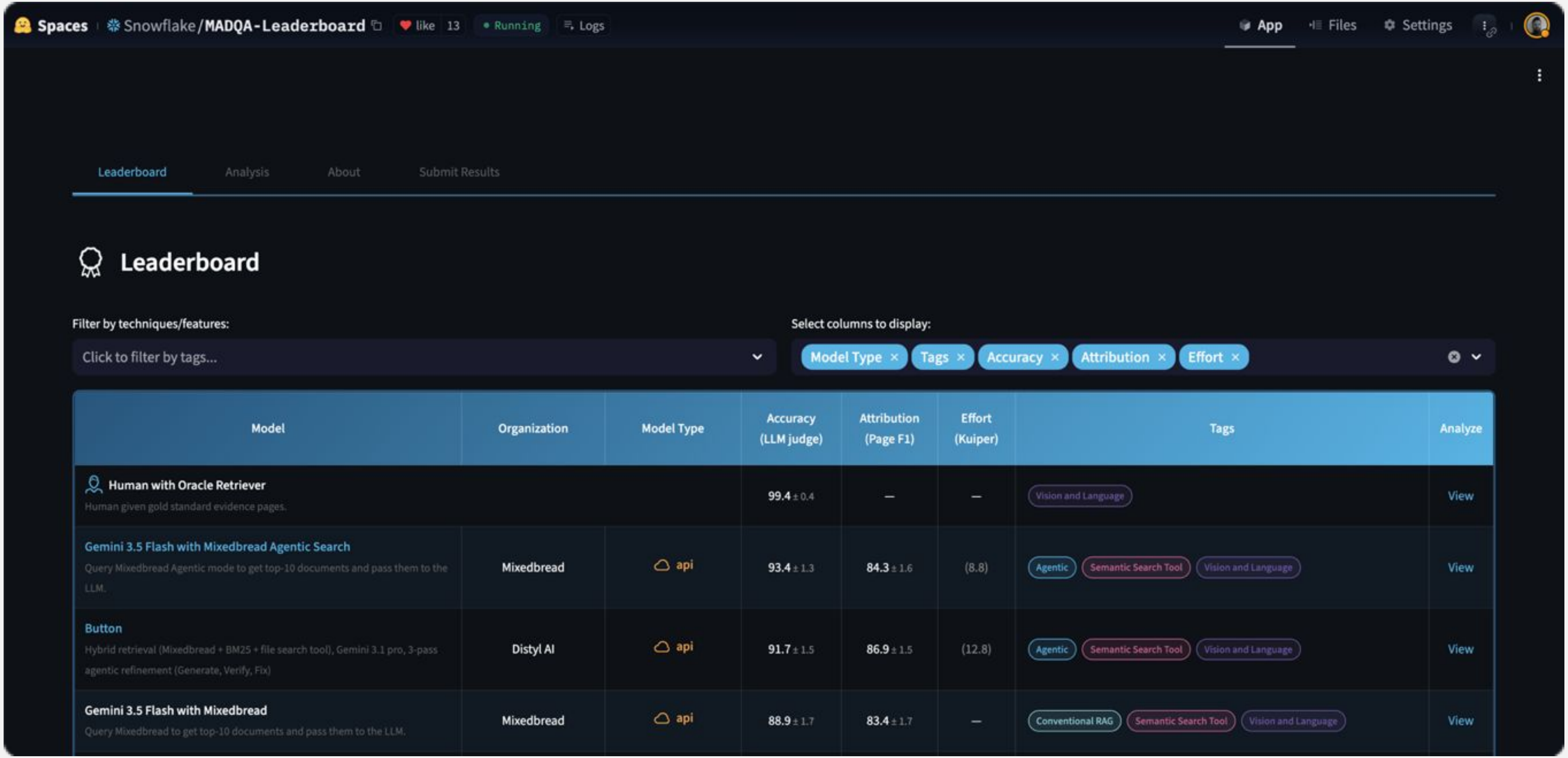
2015 Ranking

Key: Percent of total points available attained by state: 80-100 | 60-80 | 40-60 | 20-40 | 0-20

STATE	2015 Rank	2014 Rank	2015 Points	2014 Points	Legislation & Enforcement	Policies & Programs	Infrastructure & Funding	Education & Encouragement	Evaluation & Planning
Washington	1	1	66.2	66.8					
Minnesota	2	2	62.7	62.0					
Delaware	3	4	54.8	55.7					
Massachusetts	4	10	54.8	53.7					
Utah	5	8	54.3	53.7					
Oregon	6	5	54.2	55.2					
Colorado	7	6	53.9	54.1					
California	8	9	53.1	53.7					
Wisconsin	9	3	52.2	56.9					
Maryland	10	7	49.0	53.8					
New Jersey	11	12	48.6	53.0					
Pennsylvania	12	19	47.9	41.4					
Virginia	13	18	47.2	41.5					
Illinois	14	11	46.0	53.1					
Maine	15	13	45.6	50.6					
Ohio	16	16	45.3	45.1					
Vermont	17	17	43.3	44.7					
Michigan	18	14	42.8	50.1					
Arizona	19	15	42.2	46.7					
Tennessee	20	22	42.0	39.7					
Idaho	21	20	41.7	41.1					
Connecticut	22	21	41.4	40.0					
North Carolina	23	23	39.1	39.5					
Florida	24	28	38.7	35.3					
Georgia	25	26	37.5	38.6					
Rhode Island	26	27	36.1	38.5					
New Hampshire	27	24	35.9	38.7					
Iowa	28	25	35.7	38.6					
New York	29	29	35.4	33.9					
Texas	30	33	35.2	31.0					
Nevada	31	30	35.1	33.8					
Mississippi	32	31	34.5	32.8					

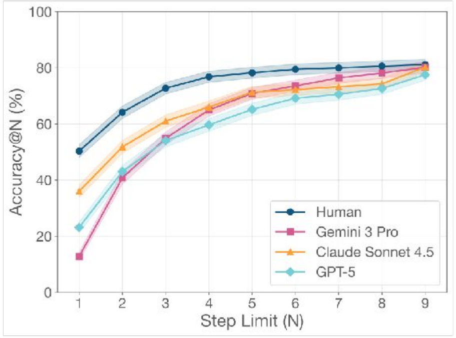
Leaderboard

https://huggingface.co/spaces/Snowflake/MADQA-Leaderboard



Model	Organization	Model Type	Accuracy (LLM judge)	Attribution (Page F1)	Effort (Kuiper)	Tags	Analyze
Human with Oracle Retriever Human given gold standard evidence pages.			99.4 ± 0.4	–	–	Vision and Language	View
Gemini 3.5 Flash with Mixedbread Agentic Search Query Mixedbread Agentic mode to get top-10 documents and pass them to the LLM.	Mixedbread	api	93.4 ± 1.3	84.3 ± 1.6	(8.8)	Agentic, Semantic Search Tool, Vision and Language	View
Button Hybrid retrieval (Mixedbread + BM25 + file search tool), Gemini 3.1 pro, 3-pass agentic refinement (Generate, Verify, Fix)	Distyl AI	api	91.7 ± 1.5	86.9 ± 1.5	(12.8)	Agentic, Semantic Search Tool, Vision and Language	View
Gemini 3.5 Flash with Mixedbread Query Mixedbread to get top-10 documents and pass them to the LLM.	Mixedbread	api	88.9 ± 1.7	83.4 ± 1.7	–	Conventional RAG, Semantic Search Tool, Vision and Language	View

The best LLM agents can now match human accuracy in document intelligence tasks.



But they solve problems 5 times less efficiently.

And both humans and machines hit a performance ceiling.

Correctness

Is the answer correct and actionable?

Attribution

Is the answer well grounded?

Efficiency

Is the effort the agent invested justified?

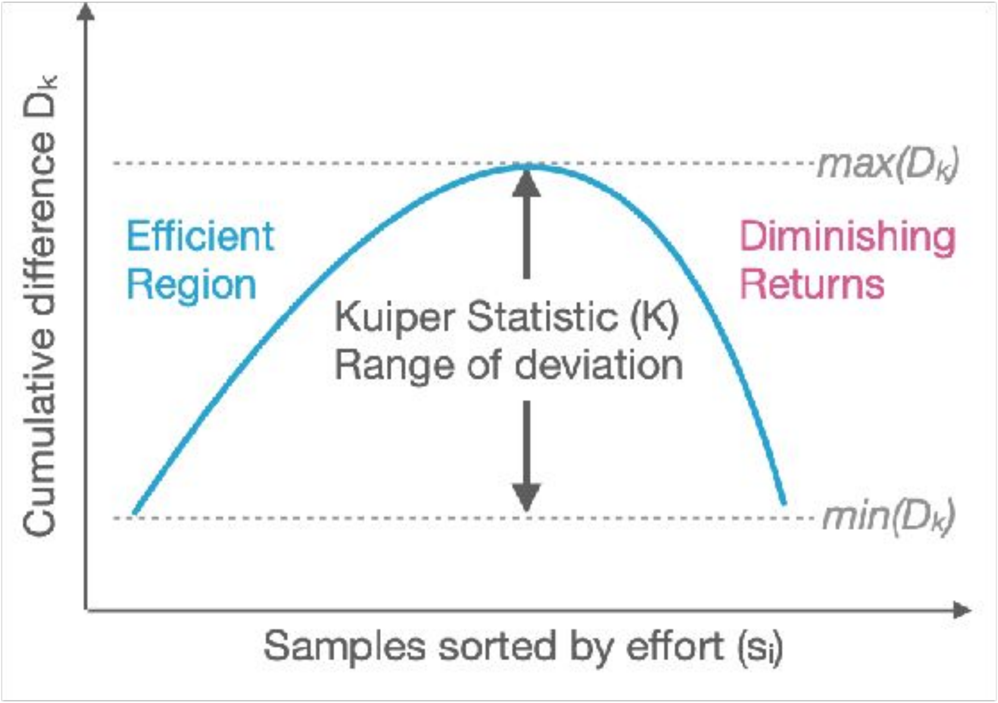
Findings

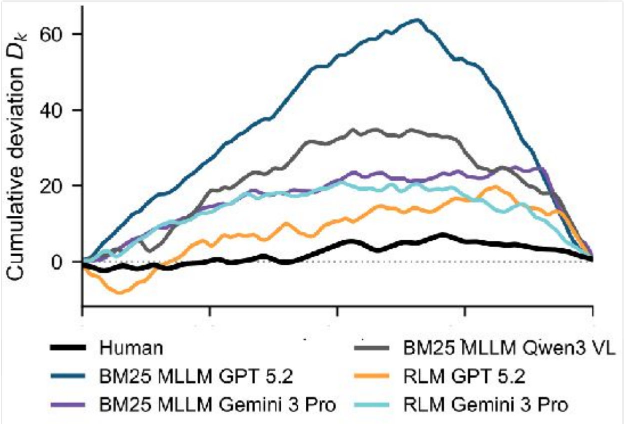
Simple Agentic Systems Can Outperform Strong, Static RAG, e.g.:

	Static RAG Service	Acc.	BM25 Agent Acc.	Δ
Gemini	*Gemini File Search*	78.6	82.2	**+3.6**
GPT	*OpenAI Assistants File Search*	50.0	67.8	**+17.8**

Model / Framework	Accuracy
Non-Agentic Systems	
Gemini 3 Pro File Search	**78.6** ± 2.2
Gemini 2.5 Flash File Search	71.8 ± 2.4
M3DocRAG	61.6 ± 2.6
GPT-5.2 (2024-08) HEAVEN	52.9 ± 2.7
GPT-5.2 (2025-12) File Search	50.0 ± 2.7
GPT-5 (2025-08) File Search	49.6 ± 2.7
GPT-4o (2024-08) HEAVEN	48.6 ± 2.7
GPT-5 Mini (2025-08) File Search	48.5 ± 2.7
ColBERTv2 + Llama-3.1-8B	40.2 ± 2.6
Agentic Systems	
Gemini 3 Pro BM25 Agent	**82.2** ± 2.0
Claude Sonnet 4.5 (2025-09) BM25 Agent	80.6 ± 2.1
GPT-5 (2025-08) BM25 Agent	77.7 ± 2.2
Gemini 3 Pro RLM	73.8 ± 2.3
Claude Agent Semtools	72.6 ± 2.4
Claude 4.5 Sonnet (2025-09) RLM	70.5 ± 2.4
Claude Haiku 4.5 (2025-10) BM25 Agent	68.2 ± 2.5
GPT-5.2 (2025-12) BM25 Agent	67.8 ± 2.5
GPT-5 Mini (2025-08) BM25 Agent	66.9 ± 2.5
GLM-4.6V BM25 Agent	66.1 ± 2.5
GPT-5.2 (2025-12) RLM	64.2 ± 2.6
MDocAgent	63.8 ± 2.6
Qwen3-VL (235B-A22B-Thinking) BM25 Agent	60.3 ± 2.6
GPT-4.1 (2025-04) BM25 Agent	60.0 ± 2.6
Gemini 2.5 Flash BM25 Agent	58.5 ± 2.6
GPT-5 Nano (2025-08) BM25 Agent	58.2 ± 2.6
Qwen3-VL (8B-Thinking) BM25 Agent	47.3 ± 2.7
GLM-4.6V Flash BM25 Agent	46.0 ± 2.7
GPT-4.1 Nano (2025-04) BM25 Agent	19.5 ± 2.1
Human Performance	
Human Oracle Retriever	**99.4** ± 0.4
Human BM25 Agent	82.2 ± 2.0

Adaptive compute: effort calibration?





Section 02

Inner Loop

EVIDENCE

Can I trust it?

EVIDENCE · VISUAL COMPLEX RECOGNITION

Can I trust what I found?

Provenance Crop-Based Agentic Reflection

BASELINE PREDICTION

2, 5, 6, 9, 11, 12, 14, 15, 18, 19, 20 ❌

IDP-AGENT PREDICTION

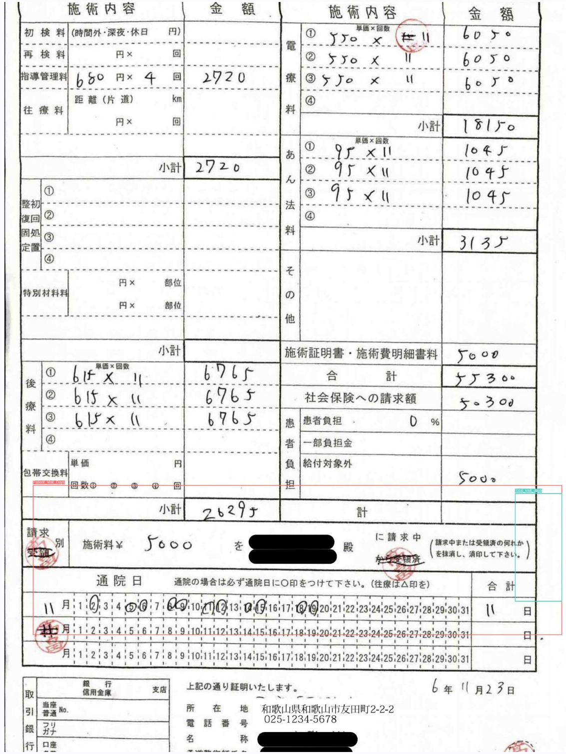
2, 5, 6, 8, 9, 11, 12, 14, 15, 18, 19 ✅

"The corrected list matches the 'Total' (合計) count of 11 provided in the image by resolving circled marks."

Agent resolved complex handwriting

First-pass extraction → **Provenance / location** → **Crop** → **Inspect and verify**

Find smallest context that can test the claim.
Reflection turns evidence into automation (or trust).


施術内容 金額 施術内容 金額
初検料 (時間外・深夜・休日 円)
再検料 円× 回
指導管理料 680 円× 4 回 2720
往療料 距離(片道) km 円× 回
小計 2720
電療料 ① 550 × 11 6050 ② 550 × 11 6050 ③ 550 × 11 6050 ④ 小計 18150
あん法料 ① 95 × 11 1045 ② 95 × 11 1045 ③ 95 × 11 1045 ④ 小計 3135
整復固定 初回処置 ① ② ③ ④
特別材料料 円× 部位 円× 部位
その他
小計
施術証明書・施術費明細書料 5000
後療料 ① 615 × 11 6765 ② 615 × 11 6765 ③ 615 × 11 6765 ④
合計 55300
社会保険への請求額 50300
患者負担 患者負担 0 % 一部負担金 給付対象外 5000
包帯交換料 単価 円 回数
小計 26295 計
請求別 施術料¥ 5000 を [redacted] 殿 に請求中
通院日 通院の場合は必ず通院日に○印をつけて下さい。(往療は△印を) 合計
11 月 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 11 日
月 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 日
月 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 日
取引銀行 銀行 信用金庫 支店 当座 普通 No. フリガナ 口座
上記の通り証明いたします。 6 年 11 月 23 日
所在地 和歌山県和歌山市友田町2-2-2
電話番号 025-1234-5678
名称 [redacted]

PHENOMENA OF INTEREST: Handwriting Recognition (HWR) • Small/Fine Print in Forms • Non-Latin Scripts • Visually Complex Reasoning (checkboxes, circles, cross-outs)

Section 02

Inner Loop

ACTION

What should I do?

What should I do?

ROUTING ENGINE

Dynamic Compute Allocation

Once evidence is gathered and verified, the system must decide what should happen next and what entity should perform it. The agent routes the action to the most appropriate substrate.

01 Model Classes

LLMs, VLMs, and specialized models selected dynamically for reasoning and unstructured processing.

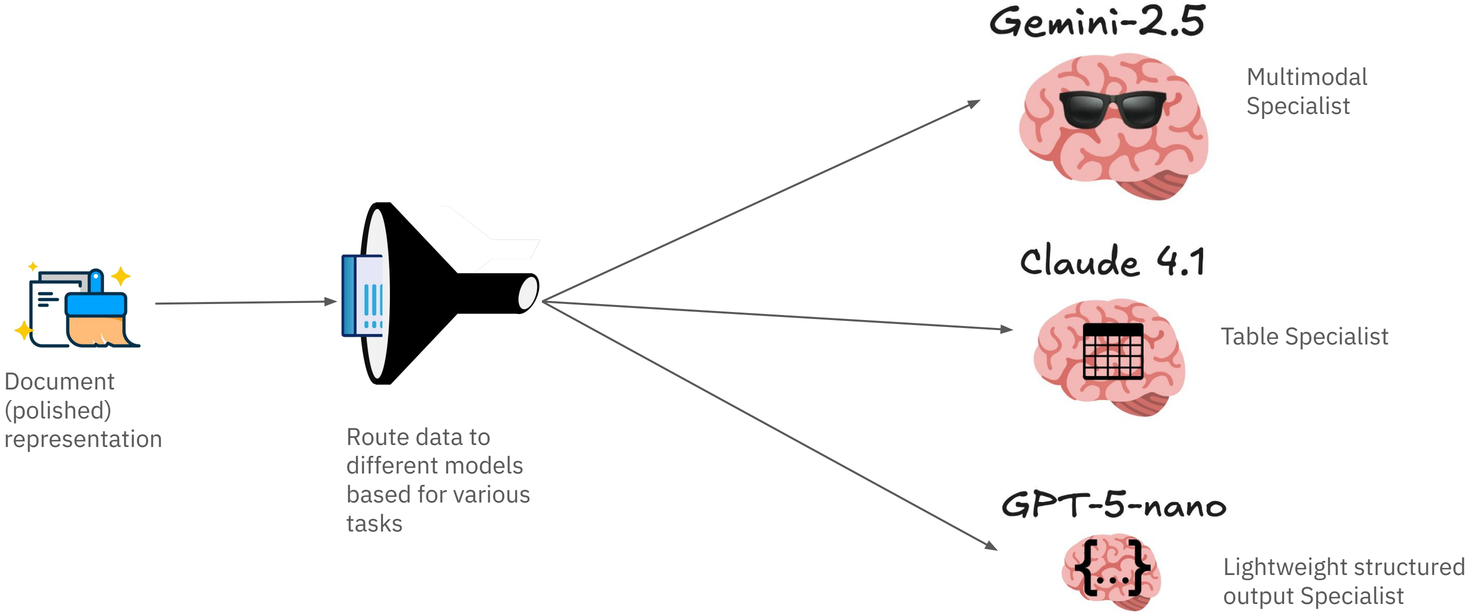
02 Deterministic Workflows

Parsers and predefined logic paths utilized when execution must remain strictly bounded and reliable.

03 Code & Human Execution

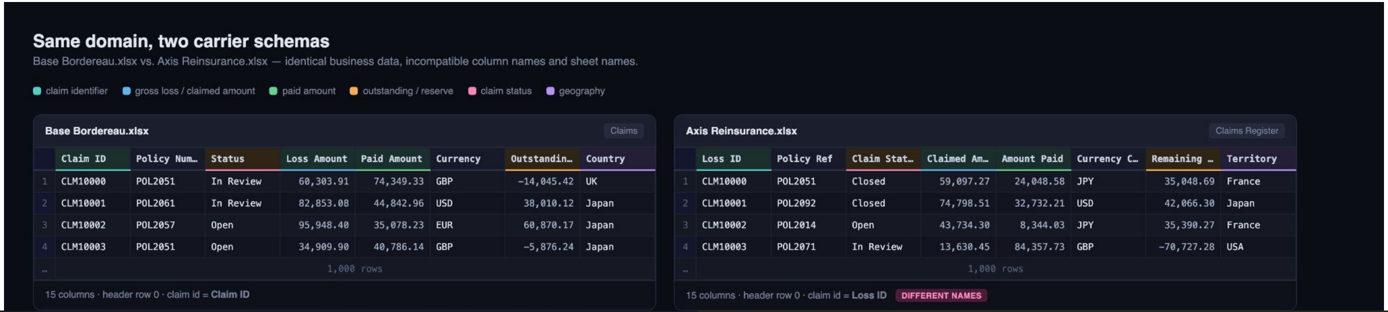
Python playbooks, direct SQL queries, or escalation to human experts when high-fidelity execution is required.

Example 1: Routing according to complexity/cost-effectiveness



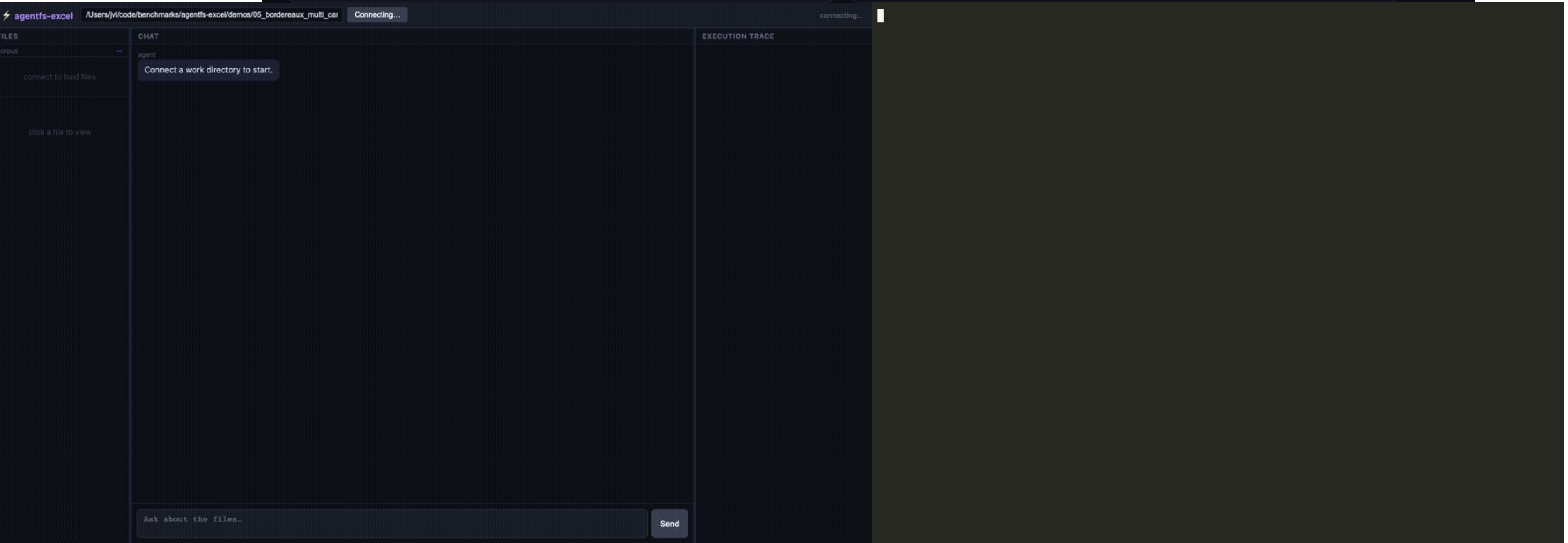
INSTABASE

Example 2: structured data reconciliation

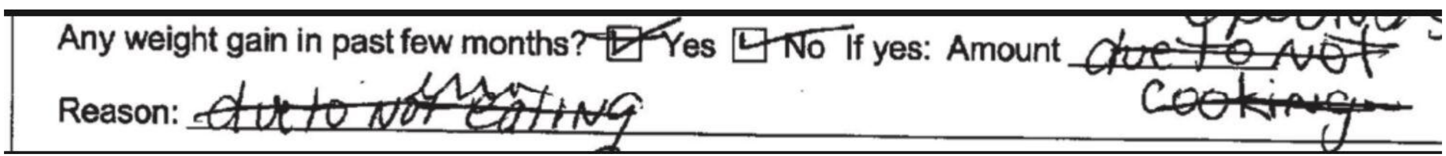


	Claim ID	Policy Num…	Status	Loss Amount	Paid Amount	Currency	Outstandin…	Country
1	CLM10000	POL2051	In Review	60,303.91	74,349.33	GBP	-14,045.42	UK
2	CLM10001	POL2061	In Review	82,853.08	44,842.96	USD	38,010.12	Japan
3	CLM10002	POL2057	Open	95,948.40	35,078.23	EUR	60,870.17	Japan
4	CLM10003	POL2051	Open	34,909.90	40,786.14	GBP	-5,876.24	Japan
…	1,000 rows							

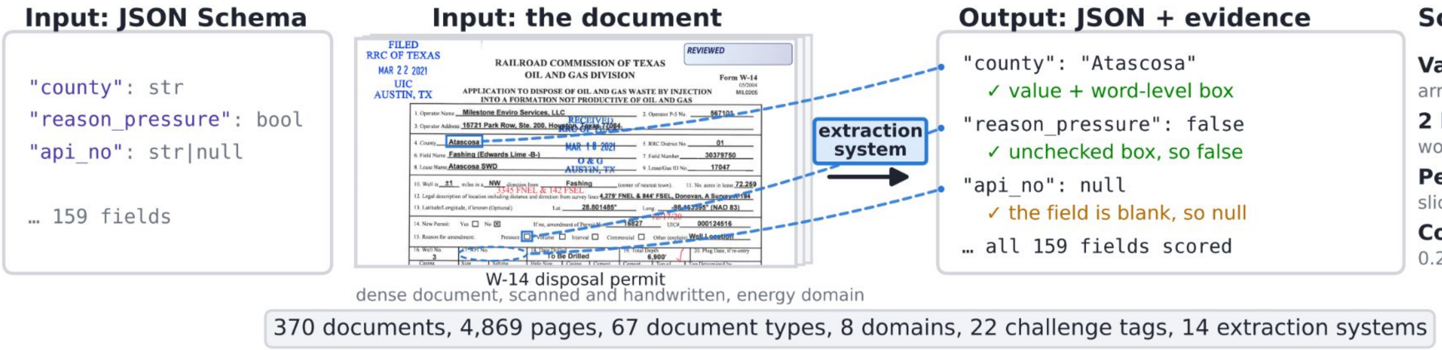
	Loss ID	Policy Ref	Claim Stat…	Claimed Am…	Amount Paid	Currency C…	Remaining …	Territory
1	CLM10000	POL2051	Closed	59,097.27	24,048.58	JPY	35,048.69	France
2	CLM10001	POL2092	Closed	74,798.51	32,732.21	USD	42,066.30	Japan
3	CLM10002	POL2014	Open	43,734.30	8,344.03	JPY	35,390.27	France
4	CLM10003	POL2071	In Review	13,630.45	84,357.73	GBP	-70,727.28	USA
…	1,000 rows							

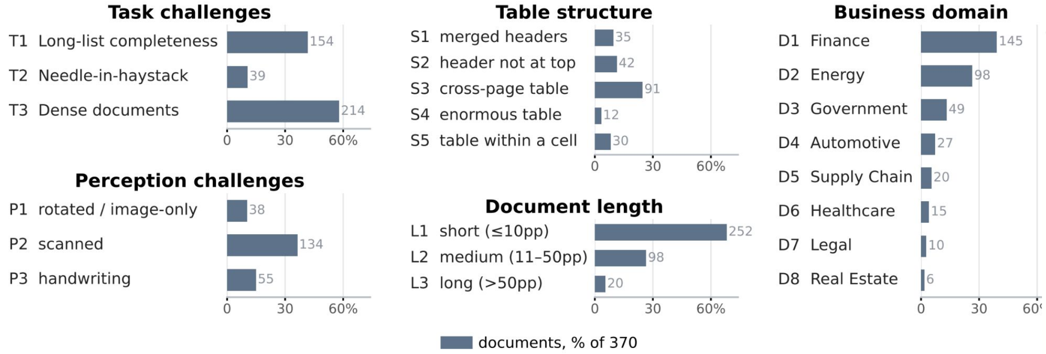


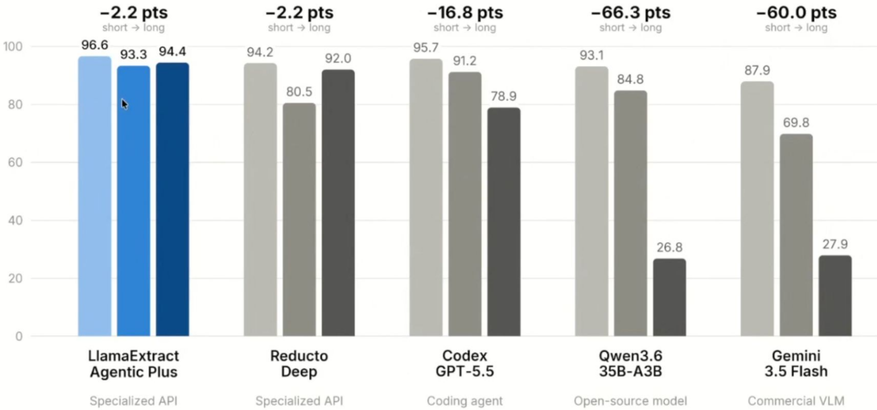
Example 3: boundary condition of human escalation


Any weight gain in past few months? ☑ Yes ☐ No If yes: Amount ~~due to not~~
cooking
Reason: ~~due to not eating~~

ExtractBench







FEEDBACK LOOP — THE SYSTEM LEARNS

Section 03

Outer Loop

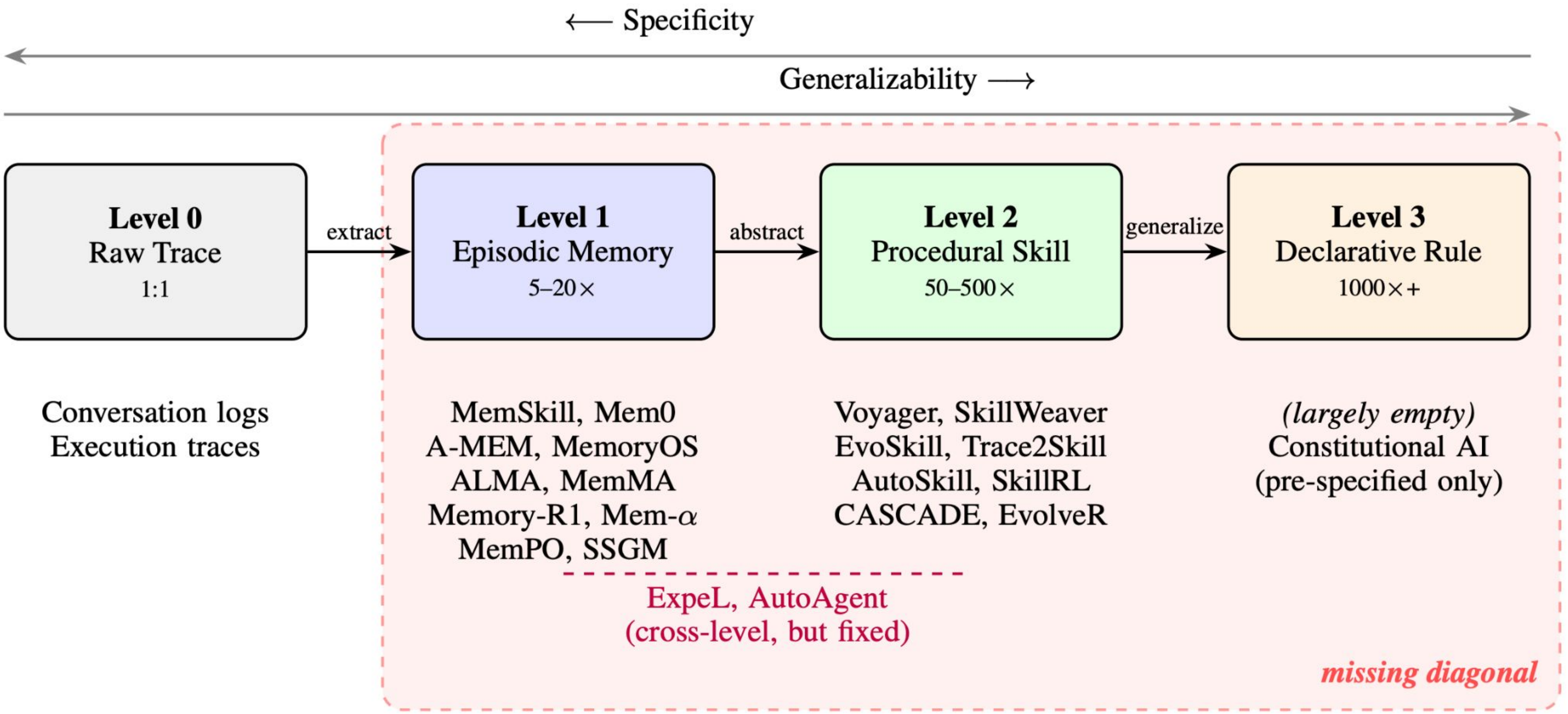
Experience → Feedback → Change the system

Remember

Promote

Compile

Experience compression spectrum (Zhang et al. 2026)



Don’t waste experience

SUCCESS | FAILURE | HUMAN CORRECTION | ESCALATION | TOOL TRACE | PRODUCTION DRIFT

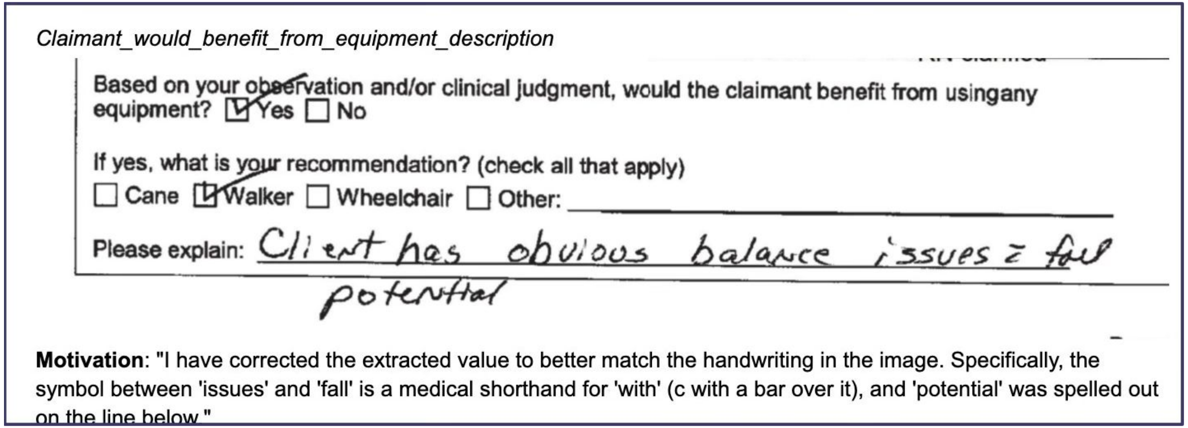
GUIDING PRINCIPLE

Missing competence
→ scaffold it.

Stable competence
→ compile it.

Residual uncertainty
→ keep it adaptive.

Walkthrough example (**success**)



Claimant_would_benefit_from_equipment_description

Based on your observation and/or clinical judgment, would the claimant benefit from usingany equipment? ☑ Yes ☐ No

If yes, what is your recommendation? (check all that apply)
☐ Cane ☑ Walker ☐ Wheelchair ☐ Other: ____

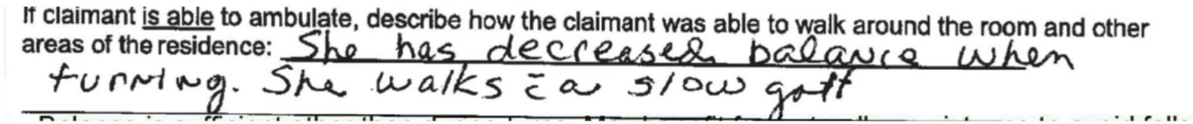
Please explain: Client has obvious balance issues c̄ fall potential

Motivation: "I have corrected the extracted value to better match the handwriting in the image. Specifically, the symbol between 'issues' and 'fall' is a medical shorthand for 'with' (c with a bar over it), and 'potential' was spelled out on the line below."

handwriting on evidence page in medical form → provenance reflection

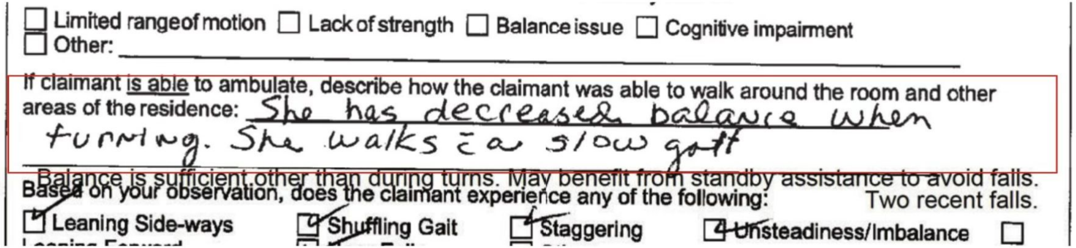
Walkthrough example (**failure**)

Box too tight



If claimant is able to ambulate, describe how the claimant was able to walk around the room and other areas of the residence: She has decreased balance when turning. She walks c̄ a slow gait

→



☐ Limited rangeof motion ☐ Lack of strength ☐ Balance issue ☐ Cognitive impairment
☐ Other:

If claimant is able to ambulate, describe how the claimant was able to walk around the room and other areas of the residence: She has decreased balance when turning. She walks c̄ a slow gait

Balance is sufficient other than during turns. May benefit from standby assistance to avoid falls.
Based on your observation, does the claimant experience any of the following: Two recent falls.

☐ Leaning Side-ways ☐ Shuffling Gait ☐ Staggering ☐ Unsteadiness/Imbalance ☐

handwriting on evidence page in medical form → provenance region*150%

Walkthrough example (**failure**)

Medical Condition	Onset Date (MM / DD / YYYY)
Diabetes	1 / 1 / 2019
Vit B12 deficient	1 / 1 / 2020
Balance Issues	1 / 1 / 2022
Vit D3 deficient	1 / 1 / 2022
Fall potential	1 / 1 / 2022
Wt loss	1 / 1 / 2023
Dementia	Unknown /
Hypertension	Unknown /
	/ /
	/ /
	/ /
	/ /

Claimant-Reported: Height 5 ft. 2 in. Weight ~~104~~ 102 lbs.

Any weight gain in past few months? ☐ ~~Yes~~ ☐ ~~No~~ If yes: Amount ~~8 pounds due to not cooking~~

Reason: ~~due to not eating~~

Any weight loss in past few months? ☑ Yes ☐ No If yes: Amount 8 pounds

Reason: due to lack of interest in eating

Walkthrough Example: (Parametric Updates)

Train Vision-Language Models to generate better search queries through trajectory learning.

💡 Use the **normalized rank of GT doc/pages** as a reward signal to train the model to generate more effective search queries.

<u>Question</u>: Search for documents about apache helicopter specification

```
Output: <think>
Okay, the user is asking for documents about Apache helicopter specifications.
Hmm, this seems straightforward but I need to think carefully. They're probably a
researcher, military enthusiast, or maybe someone in aviation industry. Could also
be a student working on a project.

First thought: They didn't specify which Apache variant (UH-60 vs AH-64). That's
crucial because specs differ wildly. The AH-64 Apache attack helicopter has very
different specs than the UH-60 Black Hawk transport. Should clarify that
immediately to avoid misinformation.

(...)

*preparing response*
Will start with variant clarification, then pivot to search strategies. The
"search terms" part is most actionable - they'll probably copy-paste those. Should
include both exact phrases ("AH-64 Block III") and broad terms ("Apache
specifications").
```

[Think]
Okay, the user is asking for a search on Apache helicopter specifications. (...)

Apache specs usually cover things like speed, range, armament, and capabilities. I recall the AH-64E is the latest variant, but I should confirm if they want all models or just current ones. Since they didn't specify, I'll cover the main variants. (...)

Final thought: Keep it precise but offer flexibility. They might actually want maintenance manuals or flight manuals - but since they said "specifications", I'll focus on performance and capabilities first.

[Search Query]
Apache helicopter specifications AH-64E technical data armament speed range sensor systems engine details

Project Code & Implementation: https://github.com/jvlinsta/DRAG/

RESEARCH AGENDA

Start benchmarking failure modes.

REPRESENTATION

What does faithful mean for an AI-native document substrate?

EVIDENCE

Can evidence-grounded signals outperform scalar confidence on hard fields?

SEARCH

Can trajectories teach calibrated stopping and human–agent routing?

EXECUTION

Can the language-versus-code boundary be learned and audited?

METHOD **Enumerate the failure taxonomy first; synthesize one minimal case per failure.**

We don’t eliminate uncertainty.

We aim for systems that can reduce it, check it, and learn from what happens over time.

REPRESENT → SEARCH → EVIDENCE → ACT → EXPERIENCE → BETTER SYSTEM ↺

Reduce uncertainty. Don’t waste experience.

Questions?

References

Benchmarks & Datasets

- **DUDE** (still not solved!)
- **MADQA** (arXiv:2603.12180)
- **Parsebench & Extractbench** [LlamaIndex]
- **Micro1 LongExtract-50** [Reducto]
- **RealDocBench** [Extend]
- **ConfBench** (arXiv:2608.01792)

- **OfficeQA Pro v2** [DataBricks]
- **GDPVal** [OpenAI]
- **GPD.pdf** [Surge AI]
- **EnterpriseOps-Gym** [ServiceNow]

Agent Architectures & Memory

- **Experience Compression Spectrum: Unifying Memory, Skills, and Rules in LLM Agents** (arXiv:2604.15877)
- **SkillSmith: Learning to Compose Parametric Skills & Knowledge** (arXiv:2607.27497)
- **Compiled AI: Deterministic code generation for workflow automation** (Trooskens et al., 2026)
- **Inference Time Feedback for Agents** (arXiv:2604.27233)
- **SIA: Self Improving AI with Harness & Weight Updates** (arXiv:2602.07755)
- **Mem-π: Adaptive Memory through Learning When and What to Generate** (arXiv:2605.21463)
- **Learning to Continually Learn via Meta-learning Agentic Memory Designs** (arXiv:2605.27276)